April 1, 2026

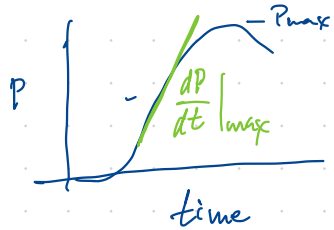
Explosion Venting:

$$K_{ST} = \left(\frac{dP}{dt}\right)_{max} \cdot V^{1/3}$$

K_{ST} for methane:

K_G 45-80 bar m/s

P_{max} = 6.8 – 8.7 bar ←



Method 1: NFPA 68

$$A_v = \frac{A_s C}{\sqrt{P_{red}}}$$

[for $P_{red} \leq 0.5$ bar-g

P_{red} - max pressure in vented enclosure (bar-g)

$C = 0.0223 \lambda S_u$ ($bar^{1/2}$, S_u in m/s)

Eq. 7.2.1.7 [P_{max} < 9 bar_g]

S_u = fundamental burning velocity

for methane & air: ≈ 0.4 m/s

λ = ratio of burning velocity for turbulent to laminar burning velocity

≈ 0.6/0.4 = 1.5 for methane/air

A_s = Internal Surface Area

C = 0.0134 $bar^{1/2}$

30 L sphere

$V = \frac{4}{3}\pi r^3$ $V = 0.19\ m$

$A_s = 4\pi r^2$

$A_s = 0.45\ m^2$

A_v = $\pi \cdot 0.076^2$ (6 in dia)

= 0.018 m^2

$$P_{red} = \left(\frac{A_s}{A_v} \cdot C\right)^2 = \left(\frac{0.45}{0.018} \cdot 0.0134\right)^2 = 0.11 \text{ bar}$$

≈ 1.5 psig

NFPA 68

during the venting process.

6.10.8.2 The restricted area shall be identified as an electrically classified (hazardous) area in accordance with *NFPA 70*.

6.10.8.3 There shall be no normally present ignition sources in the restricted area, including, but not limited to, hot surfaces exceeding the auto-ignition temperature and open flames.

6.10.9* Devices without particulate retention elements that reset after relieving pressure shall be inspected after a deflagration to ensure that the design performance has not been affected.

Chapter 7 Venting Deflagrations of Gas Mixtures and Mists

7.1 Introduction.

7.1.1* This chapter shall apply to the design of deflagration vents for enclosures that contain a flammable gas or combustible mist and that have an L/D of ≤5.

7.1.1.1 This chapter shall be used in conjunction with the information contained in the rest of this standard.

7.1.1.2 Chapter 6 and 3.3.32.1 shall be reviewed before using this chapter.

7.2 Venting by Means of Low Inertia Vent Closures.

7.2.1 Low Inertia Vent Closure Equations for Low P_{red}. When $P_{red} \le 0.5$ bar-g, the minimum required vent area, A_{v0}, shall be determined by Equation 7.2.1a and Equation 7.2.1b:

$$A_{v0} = \frac{A_s C}{\sqrt{P_{red}}} \qquad [7.2.1a]$$

$$C = \frac{S_u \rho_u \lambda}{2 G_u C_d}\left[\left(\frac{P_{max}+1}{P_0+1}\right)^{1/\gamma b} - 1\right](P_0+1)^{1/2} \qquad [7.2.1b]$$

where:

A_{v0} = the vent area calculated from Equation 7.2.1a (m^2)

A_s = the enclosure internal surface area determined according to 7.2.5 (m^2)

P_{red} = the maximum pressure developed in a vented enclosure during a vented deflagration (bar-g)

7.2.1.1 The C value for flammable gases and vapors with a value less than 9 bar-g and a stoichiometric (near worst) fuel concentration no greater than about 10 percent shall be permitted to be calculated using Equation 7.2.1.1 for Equation 7.2.1a:

$$C = 0.0223 \lambda S_u \text{ bar}^{1/2} \text{ for } S_u \text{ in m/s}$$

7.2.1.2 When applying Equation 7.2.1a, the value of P_{red} shall be less than P_{red} as specified for the following conditions:

(1) For $P_{red} \le 0.1$ bar-g (1.5 psig), $P_{stat} \le P_{red} - 0.024$ b psf).

(2) For $P_{red} > 0.1$ bar-g (1.5 psig), $P_{stat} < 0.75\ P_{red}$.

Δ 7.2.2 Low Inertia Vent Closure Equations for High P_{red}. $P_{red} > 0.5$ bar-g, the minimum required vent area, A_{v0}, determined from Equation 7.2.2a and Equation 7.2.2b:

$$A_{v0} = A_s \frac{\left[1 - \left(\frac{P_{red}+1}{P_{max}+1}\right)^{1/\gamma b}\right]}{\left[\left(\frac{P_{red}+1}{P_{max}+1}\right)^{1/\gamma b} - \delta\right]} \frac{S_u \rho_u}{G_u} \frac{\lambda}{C_d}$$

$$\delta = \frac{\left(\frac{P_{stat}+1}{P_0+1}\right)^{1/\gamma b} - 1}{\left(\frac{P_{max}+1}{P_0+1}\right)^{1/\gamma b} - 1}$$

where:

A_{v0} = the vent area calculated from Equation 7.2.2a (m

A_s = the enclosure internal surface area determined a ing to 7.2.5 (m^2)

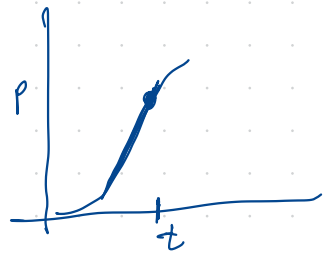
P_{red} = the maximum pressure developed in a vented en during a vented deflagration (bar-g)

S_u = fundamental burning velocity of gas-air mixture

ρ_u = mass density of unburned gas-air mixture (kg/m

Method 2: K_{ST} or K_G

$$K_G = \left.\frac{dP}{dt}\right|_{max} \cdot V^{1/3}$$



from NFPA 68

$$t = b \left(\frac{P_{max}}{P_{red}}\right)^{0.5} \cdot \frac{V}{A_v}$$

(V: volume)

$b = 4.3 \cdot 10^{-3}$ (Eq. 6.3.5.5 in NFPA 68)

For our specific scenario

$K_G = 45 - 80$ bar m/s

$V = \frac{4}{3}\pi r^3 = 0.03\ m^3$ $\quad V^{1/3} = 0.31$ m

$$\left.\frac{dP}{dt}\right|_{max} = \frac{45-80 \text{ bar}\frac{m}{s}}{0.31 \text{ m}} =$$

$$145 - 258 \ \frac{bar}{s}$$

$$t = 4.3 \cdot 10^{-3} \cdot \left(\frac{8.7 \text{ bar}}{0.2 \text{ bar}}\right)^{0.5} \cdot \frac{0.03}{0.018} = 0.047 \text{ sec}$$

+ 118%
− 37%

or

0.03 − 0.1 sec

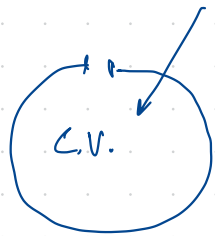
For $\left.\frac{dP}{dt}\right|_{max}$:

$P_{internal\ red}$ ~ 6.8 − 12 Bar (why? 0.1 Bar)

$= 145 \cdot 0.047$



Method 3: Mass & Energy Balance



$$\left.\frac{dU}{dt}\right|_{C.V.} = \cancel{\dot{n}_{in}} - \dot{n}_{out} + \underline{gen}$$

$\dot{n}_{out}$ – outflow through the vent once the burst disk ruptures

gen – combustion gases

$$CH_4 + 2O_2 + _\, N_2 \rightarrow CO_2 + 2H_2O + _\, N_2$$

$$3 + __ \rightarrow 3 + __$$

No change in moles

$= 0$

only after burst disk functions

$$\left.\frac{d(nU)}{dt}\right|_{CV} = \cancel{\dot{n}_{in} H_{in}} - \dot{n}_{out}\bar{H}_{out} + \underline{gen}$$

$$gen = \Delta\hat{H}_{comb} \cdot S_u \quad [=] \quad \frac{J}{s}$$

from speed or rate of burning

P_2?

Is $P_{rel} < P_{MAWP}$

Burn velocity in m/s $\rightarrow$ $k = Ae^{-E/(RT)} \cdot P^n$

then estimate $\frac{kg}{s}$ for S_u

Texas City Disaster 1947

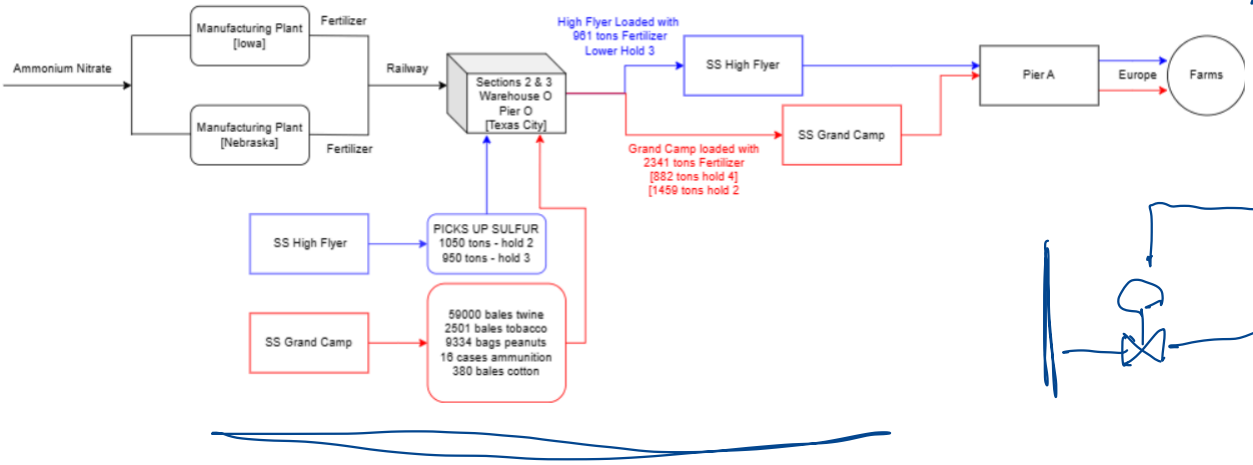
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Process Background and Scope

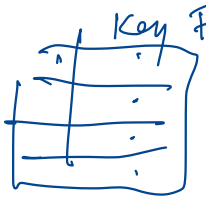
Ammonium Nitrate (NH_4NO_3) is a chemical typically used in fertilizers, however it is also commonly used in explosives as it is very sensitive to large quantities of heat and pressure. After World War II, many European countries were working to rebuild themselves. The United States helped by manufacturing and shipping fertilizer overseas to help with farming and agricultural efforts. In 1947, the *SS Grandcamp* and the *SS Highflyer* were in Texas City, Texas picking up several tons of fertilizer when disaster struck, and an enormous explosion occurred, due to the Ammonium Nitrate on board, killing hundreds of people. It is now known as the deadliest industrial accident in United States history. In our analysis of the accident, we will discuss process safety information (PSI) and the hazards involved, beginning with the manufacturing of the fertilizer to the time that it exploded on the ships.

Process Flow Diagram

Below is a diagram of the process that the Ammonium Nitrate, the main cause of the explosion would have taken had the Texas City disaster not occurred. As shown, Ammonium Nitrate was one of the chemicals used in the manufacturing of fertilizer. Manufactures then shipped the fertilizer by railway to a warehouse in Texas City where the *SS Highflyer* and the *SS Grandcamp* picked it up. The product was to be then shipped to Europe where it would have been used on the farms there. Unfortunately, while the fertilizer was being put on the ships in Texas City, the disaster occurred, and it never made its way to Europe.



Incident Description



List of Incident

1-2 Paragraphs to discuss

At 8:00 a.m. on April 16, 1947, the SS. Grandcamp was docked in the Texas City Port holding around 2300 tons of ammonium nitrate in 100-pound paper bags, mixed with rosin. The longshoremen smelled smoke in the ship, however its source was unknown. An attempt was made to put out the fire with some water and a fire extinguisher but to no avail, and the fire department was dispatched to the dock. By the time they arrived about 45 minutes later, the ship was heating up very quickly and water was no longer able to extinguish the flames. In an attempt to smother the fire without damaging the cargo, the ship's captain decided to inert the hull with steam and close the hatch. Inerting with steam had the opposite effect, and the vapors liquefied the Ammonium Nitrate, increasing its rate of decomposition and heating the internal compartments of the ship even more. The temperature and pressure of the ship quickly approached denotation conditions, and at 9:12 a.m. the *SS. Grandcamp* exploded.

The blast produced a fifteen-foot wave, shattered windows as far as 40 miles away, leveled almost all of Texas City, and killed hundreds of people instantly. Glowing hot shrapnel hit on-shore chemical plants and refineries causing secondary fires and explosions which burned for hours. Quickly after the explosion, rescue workers from all over the country came to the city's aid, putting out fires, tending to the injured, and counting the dead.

There was another explosion at 1:10 am on April 17th on the SS. High Flyer. The SS. High Flyer, although not heavily damaged during the Grandcamp's explosion, did catch fire amidst the chaos. Filled with about 1000 tons of ammonium nitrate and 2000 tons of sulfur, this was a very volatile mixture and it was certain for explosion. However, this explosion was anticipated, and by the time of the explosion most people had left the area safely leaving the death toll to just two.

In the aftermath of both explosions 468 were confirmed dead, 113 people were never found and 5000 people were injured. 500 homes along with the entire seaport, local businesses, and over 1100 vehicles were destroyed. This disaster is considered the deadliest industrial accident in U.S history and is one of history's largest non-nuclear explosions.

Incident Contributing Factors

- Manufacturing Ammonium Nitrate fertilizer in paper bags with petroleum and paraffin wax added an unexpected fuel source
- Fertilizer was stored in a warehouse at high temperatures increasing its chemical reactivity
- On the ship, the fertilizer was stored with other flammable goods such as twine, peanuts, tobacco, and small arms ammunition which caused the fire to spread quickly
- Ammonium Nitrate fertilizer was stored in large quantities (up to 2300 tons on the *SS Grandcamp*) which increased the size of the explosion when it did occur

- The captain of the ship, in attempt to save the product, prohibited using water to stop the fire from spreading and instead tried to have the fire snuffed out
- When the fire started, using steam and closing the hatch on the ship increased the heat and pressure which ultimately caused the large explosion

Process Safety Information

Ammonium Nitrate

Physical and Chemical Properties

Ammonium Nitrate is a white, odorless, solid that acts as an oxidizer and an irritant. It will not burn, but it can cause fire when it comes into contact with combustible and organic material. Its decomposition temperature is 210 °C and its solubility is 190 g/100 mL at 20°C.

Stability and Reactivity

This substance is a reactive hazard and has an explosion risk when exposed to fire. Additionally, it is hygroscopic, meaning it can absorb moisture from the air. It is stable under normal conditions. Despite having an explosion risk, the substance is not sensitive to mechanical impact or static discharge. But it is incompatible with strong oxidizing agents, strong reducing agents, strong acids, finely powdered metals, and combustible materials. This compound undergoes thermal decomposition in the presence of heat. This reaction produces irritating gasses and vapors such as nitrogen oxides (NOx) and ammonia.

Toxicological and Ecological Information

Ammonium nitrate can be irritating to the eyes and lungs. Its LC50 inhalation measurement is 88.8 g/L for a rat over four hours. This material should not be emptied into drains and any releases to the environment should be avoided. It can be toxic to fish with a measured freshwater fish LC50 of 447 mg/L over 48 hours.

Handling and Storage

When storing, care should be used to ensure it is not mixed with combustibles or stored near them. The storage area should be cool and well-ventilated. Avoid the presence of moist air and water. Precautions to eliminate heat and ignition sources should be taken, such as prohibiting smoking. Since this dust can be explosive, care should be taken to avoid dust accumulation and ensure that the container is tightly closed. This material may be corrosive to metals, so be sure to store it in a container made of an appropriate noncombustible, nonmetal container.

Exposure Controls and Personal Protection

Appropriate personal protection equipment should be worn while handling ammonium nitrate. This includes protective gloves, protective clothing, and eye/face protection. Any exposed skin should be washed after handling. If any material gets in the eyes, rinse cautiously with water for several minutes. In the

event of a fire involving ammonium nitrate, evacuate the area and fight the fire remotely. This precaution is due to the risk of explosion/detonation of the ammonium nitrate. To minimize injury, ensure that people evacuate one mile or more away from the fire. Firefighters should water self-contained breathing apparatuses and may use general extinguishers any initial fire that does not involve ammonium nitrate.

Sulfur

Physical and Chemical Properties

Sulfur is a yellow solid that is odorless in its pure form but smells of rotten egg if it contains trace amounts of H2S. This material acts as an irritant, flammable solid, and combustible dust. This compound is insoluble in water. It has a lower explosion limit of 3.3% by volume and an upper explosion limit of 46% by volume. The flash point occurs at 207 °C and will auto-ignite once it reaches 232 °C.

Stability and Reactivity

This substance is generally stable and nonreactive under normal conditions. In the presence of heat, it can undergo a thermal decomposition reaction that releases irritating gasses and vapors such as sulfur oxides (SOx). This dust can create an explosion if ignited by static discharge. Grinding sulfur may produce an additional explosion hazard. Its lower flammability limit is 35 g/m3 and its upper flammability limit is 1400 g/m3. Sulfur will reach its flash point at 207 °C and its autoignition temperature at 232 °C.

Toxicological and Ecological Information

Prolonged contact with sulfur dust may cause irritation. Continued prolonged inhalation over several years may even cause respiratory disease. The LC50 by inhalation for a rat over four hours is 9.23 mg/L. This substance should not be released into the environment and has a LC50 of 180 mg/L over 96 hours for Oncorhynchus mykiss (rainbow trout). Accidental releases environmental precautions to protect bodies of water include diking and absorbents.

Handling and Storage

Sulfur should be stored in tightly closed containers away from heat and ignition sources. To help avoid ignition specifically by static discharge, the storage container and receiving equipment should be grounded or bonded. Explosion-proof electrical, ventilation, and lighting equipment should be used. All spills should be cleaned up immediately with a non-sparking tool. To avoid dispersing dust into the air, use a vacuum with a HEPA filter rather than compressed air to clear dust piles. This material should not be pressurized, cut, welded, brazed, soldered, drilled, or ground. Ensure the material is in a cool, dry, and well-ventilated area. Some additional safeguards to minimize hazards are prohibiting smoking in the storage area and locking the area when not in use.

Exposure Controls and Personal Protection

Appropriate personal protection equipment should be worn while handling sulfur. This includes protective gloves, protective clothing, and eye/face protection. When sulfur ignites, it burns as a pale blue flame that can be difficult to observe in the daylight. The byproducts of combustion include sulfur dioxide, a suffocating gas that can cause severe lung damage. The area should be evacuated and the flames combated from a safe distance. Firefighters can use water spray, fog, or foam to extinguish the flame. The use of water spray may cause frothing of the material. The flame can reignite after it is extinguished, so it is essential to cool the containers with water long after the fire is out to ensure it does not resume burning.

Pressure

Ammonium nitrate and sulfur both have thermal decomposition reactions that generate more moles than there were initially. If this reaction occurs in an enclosed space, the pressure in the area will build until an explosion occurs. The first step to prevent this would be to ensure the material is not exposed to heat. A second level of protection would be to add appropriately sized pressure relief valves to both the storage containers and storage area. That way if there is a pressure buildup, it can be vented before it reaches high enough pressures to cause a deadly explosion. Some good options are spring operated relief valves, buckling pin relief valves, or burst disks. Another form of pressure protection is a structural vent that will direct the explosion in one direction and protect the mechanical integrity of the rest of the storage area if an explosion occurs.

Previous Incidents

A similar explosion involving ammonium sulfate nitrate (ASN) occurred on September 21, 1921 in Oppau, Germany. It killed 507 people and injured 1917 others. The ASN was stored in large wooden silos. When the explosion occurred, there were 4,500 tons of ASN inside. The explosion occurred due to the use of small explosives to loosen the solidified salt created by the process. This operation had been conducted thousands of times without incident. When the company changed the drying process, it also changed the products reactivity, but they did not re-evaluate the procedure of using explosives. This incident highlights the explosive hazard associated with the storage and manufacturing of ammonium nitrate, especially in large quantities.

Recommendations

- Ammonium Nitrate should be stored at cooler temperatures to decrease chance of decomposition
- Due to flammability of the chemical, store in enclosed and undamaged containers to prevent mixing of other unwanted or flammable materials with the Ammonium Nitrate
- Store and transport in smaller quantities due to reactivity of Ammonium Nitrate
- Provide adequate ventilation in storing area to prevent an increase in pressure and temperature to decrease chemical reactivity of the chemical

- To prevent electrostatic discharge of the sulfur which causes explosions be more cautious when loading the sulfur packages and do not store damaged bags which may increase sulfur dust in the air
- Hot work should not be done on the boat while around the Ammonium Nitrate because it can cause sparks needed for it to explode

Hazards Analysis

Below are a select few of the most significant hazards from the Texas City disaster. A fault tree analysis (FTA) and a full failure modes and effects analysis (FMEA) can be found in the appendix following the report. We determined that the hazards below would have been some of the most important hazards that led to the explosion.

Events	Hazard Type	Hazard Description	Consequence	Safeguards	Rank	Recommendations
Manufacturing of ammonium nitrate (petroleum and paraffin wax)	Flammable	The petroleum and paraffin wax add a fuel source to the oxidizer (NH4NO3)	Likelihood of explosive reaction if exposed to ignition source is increased	Unknown but likely in main factory	2D	Find less reactive materials for preventing water contamination. i.e. silica gel
Transportation and storage with large amounts of (NH4NO3)	Explosion	In case of fire, the large quantities would increase chances of large explosion	Likelihood of more reactive explosion increased		1C	Store and transport in smaller quantities
Loading procedures (communication)	Ignition source	Unclear instructions/ hazards i.e. smoking and associated hazards of NH4NO3 s	Harm to workers and possible ignition source	Sign in french saying no smoking	3B	Pictogram/ multi-language signs. Safety meetings
Fire fighting (Closed deck hatches)	Explosion	Closed deck hatches results in increased pressure	Increasing pressure results in greater likelihood of larger explosion	Opening hatches	1E	Including a PRV of some sort on the ship

Fire fighting (inerting with steam)	Flammable	Steam can cause increased decomposition of ammonium nitrate	Ammonium Nitrate can decompose into nitrous oxide (oxidizer) which would contribute to fire		3E	Don't use steam to stop fire; use proper fire fighting procedures

Process Impacts

Ethical: The captain of the *SS Grandcamp* had an ethical responsibility to protect the lives of both his crew and other civilians while at port picking up cargo. Ignoring this responsibility, he focused on keeping the fertilizer on board the ship safe rather than the people. When the fire started on the ship, he ordered that no water be used to put it out so his cargo wouldn't be ruined. This is the main reason that it took too long to use water to try and get rid of the constantly spreading fire. The captain was more worried about his job and being paid for delivering the fertilizer rather than potential harm to others which should have been his greatest concern.

Global: As the cargo was intended for other countries to assist in rebuilding after WWII, it not only affected the United States, but also these European countries as they would not benefit from the assistance of fertilizer in their agricultural efforts. Additionally, in an ever growing global economy, this incident highlighted the need for labels and directions in multiple languages during international transportation.

Economic: There were several economic impacts due to the disaster. More than 500 homes were destroyed and hundreds damaged leaving 2000 homeless and resulting in an estimated $100 million in property damage (over $1 billion today). With the destruction of the Texas City port, the 11th largest US port at that time, and the surrounding buildings, the cost of reconstruction was also estimated at $100 million. Fire insurance claims reached approximately $4 million ($51 million today). Along with the economic implications of rebuilding properties and industries, $500 million in oil products were burned in the accident.

Environmental: The complete destruction of the Texas City Sea Port led to many environmental impacts. The burning wreckage in the port ignited dozens of oil and chemical tanks leading to the contamination of the port and nearby waters. For days, the city of Galveston 15 miles away was covered in an oily fog from the explosion and burning oil. And toxic fumes containing Sox, NOx, Sulfur Dioxide, Sulfurous Acid, and other hydrocarbons covered the area.

Societal: Due to the accident, approximately 600 people were killed, including all but one of the 28 in the Texas City Fire Department, and more than 5000 were injured. Many sought legal compensation, and the court case, *Elizabeth Dalehite, et al. v. United States,* sought to find the US government responsible for the negligent planning which in part resulted in the accident. Congress provided compensation and relief totaling around $17 million as a consequence of the case. Another societal impact was that due to this event and other incidents involving Ammonium Nitrate, new regulations were made for the transportation of the highly explosive chemical including storage, heat, and temperature requirements.

Recommendations

We recommend allocating $20 million from the budgets of shipping companies and the Texas City government towards establishing emergency prevention and response systems, equivalent to approximately 2% of the total disaster cost (adjusted for inflation). Within this budget allocation, we advocate for the implementation of five key initiatives: (1) minimization of hazard, (2) reducing flammability and explosiveness of fertilizer, (3) training crew on hazards awareness, (4) implementing procedural controls, and (5) development of remotely controlled tugboats.

First, we recommend minimization of hazard by containing the fertilizer in smaller quantities. To mitigate risks, we suggest limiting the quantity of stored fertilizer and imposing regulations on the proximity of ships carrying fertilizer to each other.

Second, we recommend exploring options for rendering the fertilizer less prone to combustion and explosion. This could involve incorporating inerting gasses or additives into the fertilizer, as well as storing it in fire-resistant containers rather than paper bags. Additionally, ensure that the fertilizer is not mixed with flammable materials.

Third, we recommend training the captain and crew on hazards associated with cargo, so that they can make better informed choices. For example, once the Ammonium Nitrate is sufficiently ignited, experts predict an explosion is imminent, and the proper procedure is to evacuate a mile away– not fight the fire. As part of this training, the crew should be trained for a variety of emergency scenarios, including a proper evacuation plan.

Fourth, we recommend creating strict procedural controls to eliminate sources of ignition. This could include designating specific areas as 'no hot work zones', where activities such as the use of power tools and smoking are strictly prohibited to prevent sparks or fires.

Fifth, we recommend investing in the creation of remotely operated tug boats equipped to intervene in emergencies. In the event of an imminent explosion, these tug boats can be utilized to safely remove the vessel from the harbor, while the crew evacuates, minimizing loss of human life and the risk to surrounding areas.

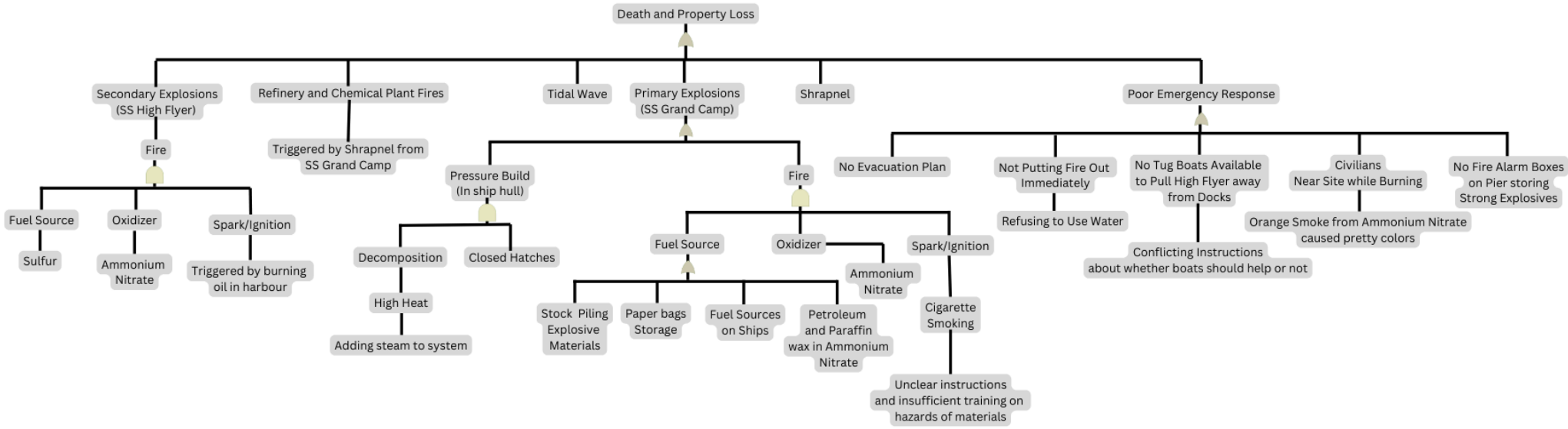
Conclusion

The Texas City Disaster of 1947 stands as a tragic reminder of the catastrophic consequences that can result from inadequate safety measures in industrial processes involving hazardous materials. The incident underscored the importance of prioritizing human safety over economic interests and highlighted the need for robust regulations and safety protocols in handling and transporting hazardous substances.

The recommendations provided offer insights into preventive measures and emergency response strategies to mitigate the risks associated with similar incidents in the future. These recommendations focus on minimizing hazards, improving communication and training, and establishing stricter procedures. It is important that the government and shipping entities work together to implement these recommendations. The lessons learned from the Texas City Disaster can be utilized to prevent future loss of human life, property, and the environment.

Appendix

Fault Tree



Failure Modes and Effects Analysis

Events	Hazard Type	Hazard Description	Consequence	Safeguards	Rank	Recommendations
Manufacturing of ammonium nitrate (petroleum and paraffin wax)	Flammable	The petroleum and paraffin wax add a fuel source to the oxidizer (NH_4NO_3)	Likelihood of explosive reaction if exposed to ignition source is increased	Unknown but likely in main factory	2D	Find less reactive materials for preventing water contamination. Ie silica gel
Manufacturing of ammonium nitrate (paper bags)	flammable	Paper bags adds a fuel source that can cause spread of fire	More likely possibility that ammonium nitrate will catch on fire		3D	Store and ship ammonium nitrate in non flammable material
Storage and transportation	Temperature Extreme	High temps during railroad transportation/ storage	High temps result in ammonium nitrate decomposition and higher chemical reactivity	Ventilation system to provide open air	3E	Store in room temp/ insulated areas
Transportation and storage with large amounts of (NH_4NO_3)	Explosion	In case of fire, the large quantities would increase chances of large explosion	Likelihood of more reactive explosion increased		1C (30 disasters since 1916)	Store and transport in smaller quantities
Shipping and Transportation	Flammable	Ammonium Nitrate was shipped with flammable materials such as ammunition and	Increased chance of fire spreading	possible separation of materials	2D	Transport materials in smaller quantities

		twine/ cotton				
Shipping and transportation	Explosion & flammable	Ignorant workers mishandle (NH_4NO_3)	High likelihood of explosion and fire loss of property and death	Unknown	1B	Provide training and SOP
Loading procedures (communication)	Ignition source	Unclear instructions/ hazards ie smoking and associated hazards of NH_4NO_3	Harm to workers and possible ignition source	Sign in french saying no smoking	3B	Pictogram/ multi-language signs. Safety meetings
Fire fighting (Closed deck hatches)	Explosion	Closed deck hatches results in increased pressure	Increasing pressure results in greater likelihood of larger explosion	Opening hatches	1E	Including a PRV of some sort on the ship
Fire fighting (inerting with steam)	Explosion/ Extreme Temperature	Using steam to fight fire results in increased both temperature and pressure	Increasing temperature/ pressure increases likelihood of larger explosion	Fire extinguishers/ fire hoses	1C	Actually use the fire hoses instead of just trying to inert with steam
Fire fighting (inerting with steam)	Flammable	Steam can cause increased decomposition of ammonium nitrate	Ammonium Nitrate can decompose into nitrous oxide (oxidizer) which would contribute to fire		3E	Don't use steam to stop fire; use proper fire fighting procedures
Sulfur burning	Toxic	Burning sulfur produces a sulfur	Corrosive damage to plant life and danger	Fire extinguisher	2D	Ship in smaller quantities, sealed

		dioxide that can react with moisture to form sulfurous acid, can explode	to humans health	water		packaging, more water to put out fire.
Burning (NH4NO3)	Explosion & flammable	Orange smoke draws a crowd towards the boat	More people closer to blast radius resulting in my injuries and dead	NA	1B	Have law enforcement make a secure perimeter
Flying shrapnel (hitting nearby refineries)	Flammable	Nearby refineries and chemical plants get hit with fast hot metal	Fire rains down on texas city	Unknown	1E	Better firefighter measures
Flying Shrapnel	Struck by	High pressure in boat hull	Boat hull burst and throws metal	Unknown	1E	PRV's to prevent pressure build
Detonation (Pressure wave)	Explosion	Huge percussion wave with large amounts of pressure	The detonation caused broken windows, leveled buildings and destroyed planes flying over	NA	1D	Prevent explosion by minimizing amount of ammonium nitrate
Detonation (Tidal wave)	Structural Damage	Large tidal wave causes leveling of several buildings	Area near the port leveled; chemicals from destroyed plants carried into the water	Unknown	2D	Build more structurally stable buildings to withstand water damage

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